

EE 115 Homework 5

- 1) **(70 points)** Assume $m(t) = \cos(2\pi t) + 2\sin(4\pi t)$ and $u(t) = (m(t) + 3)\cos(200\pi t)$ (where t is in millisecond). Also let $v(t)$ be the output of a bandpass filter (BPF) driven by the input $u(t)$. The frequency response of the BPF is assumed to be

$$H(f) = H(-f) = \begin{cases} (f - 98)/4, & 98 \leq f \leq 102 \\ 1, & 102 \leq f \leq 103 \\ 0, & \text{otherwise for } f > 0 \end{cases} \quad (1)$$

where f is in kHz.

- a) Sketch the spectrum $M(f)$ of $m(t)$.
 - b) Sketch the spectrum $U(f)$ of $u(t)$.
 - c) Sketch the spectrum $V(f)$ of $v(t)$.
 - d) Sketch the spectrum of the complex envelope of the impulse response $h(t)$ of the BPF with respect to the carrier frequency at 100kHz.
 - e) Let $v(t) = v_c(t)\cos(200\pi t) - v_s(t)\sin(200\pi t)$. Sketch the spectra of $v_c(t)$ and $v_s(t)$ separately.
 - f) How does $v_c(t)$ compare with $m(t)$?
 - g) Let $h(t) = h_c(t)\cos(200\pi t) - h_s(t)\sin(200\pi t)$. Sketch the spectra of $h_c(t)$ and $h_s(t)$ separately.
- 2) **(30 points)** To generate an FM signal $u(t) = A_c \cos(2\pi f_c + \theta(t))$ where $\theta(t) = 2\pi k_f \int_{-\infty}^t m(\tau) d\tau$, we need an integrator. If the spectrum of the message $m(t)$ has the lowest frequency at $f_L > 0$, we can use a RC circuit to implement an integrator. The transfer function of a RC circuit is

$$H(s) = \frac{\frac{1}{sC}}{R + \frac{1}{sC}}. \quad (2)$$

What is the condition on R , C and f_L such that the output of the RC circuit driven by $m(t)$ is effectively equal to $\frac{1}{RC} \int_{-\infty}^t m(\tau) d\tau$? Hint: The frequency response of the RC circuit is $H(j\omega) = \frac{1}{1 + \frac{1}{j\omega RC}}$. How to make $H(j\omega) \approx \frac{1}{j\omega RC}$?